In [1]:
```
data={
    'Age':[22,25,30,35,40,45,50,28,32,38],
    'Salary':[25000,30000,40000,50000,60000,70000,80000,90000,450000,55000],
    'Experience':[1,2,4,7,10,15,20,3,5,8],
    'Purchased':[0,0,0,1,1,1,1,0,1,1]
}
```

In [3]:
```
import pandas as pd
import numpy as np
```

In [7]:
```
df=pd.DataFrame(data)
```

In [9]:
```
df
```

Out[9]:

	Age	Salary	Experience	Purchased
0	22	25000	1	0
1	25	30000	2	0
2	30	40000	4	0
3	35	50000	7	1
4	40	60000	10	1
5	45	70000	15	1
6	50	80000	20	1
7	28	90000	3	0
8	32	450000	5	1
9	38	55000	8	1

Target variable

In [16]:
```
Y=df[['Purchased']]
X=df.drop(labels=['Purchased'],axis=1)
```

In [18]:
```
Y
```

Out[18]:

	Purchased
0	0
1	0
2	0
3	1
4	1
5	1
6	1
7	0
8	1
9	1

In [20]:
```
X
```

Out[20]:

	Age	Salary	Experience
0	22	25000	1
1	25	30000	2
2	30	40000	4
3	35	50000	7
4	40	60000	10
5	45	70000	15
6	50	80000	20
7	28	90000	3
8	32	450000	5
9	38	55000	8

Train Test split

In [23]:
```
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,Y,test_size=0.2,random_state=42)
```

Create RF

In [67]:
```
from sklearn.ensemble import RandomForestClassifier
rf=RandomForestClassifier(n_estimators=5,random_state=42)
```

Train Model

In [70]:
```
model=rf.fit(xtrain,ytrain)
```

```
C:\Users\Lenovo\anaconda3\Lib\site-packages\sklearn\base.py:1473: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
```

Prediction

In [73]:
```
y_pred=model.predict(xtest)
```

In [75]:
```
y_pred
```

Out[75]:
```
array([0, 0], dtype=int64)
```

In [77]:
```
xtest
```

Out[77]:

	Age	Salary	Experience
8	32	450000	5
1	25	30000	2

In [79]:
```
ytest
```

Out[79]:

	Purchased
8	1
1	0

Evalute Model

In [82]:
```
from sklearn.metrics import accuracy_score,confusion_matrix
accuracy=accuracy_score(ytest,y_pred)
```

In [84]:
```
accuracy
```

Out[84]:
```
0.5
```

In [89]:
```
ytest['y_pred']=y_pred
```

In [91]:
```
ytest
```

Out[91]:

	Purchased	y_pred
8	1	0
1	0	0

In [ ]:
```

```